BANKING TRANSACTION & CUSTOMER ANALYTICS

Business Conclusion Report

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August 8, 2026

Executive Summary

This report consolidates the SQL business-question analysis, exploratory data cleaning, and statistical validation performed on the bank's customer, account, transaction, and product data, and ties it to the three Power BI dashboards built on top of the same model. The dataset covers 10,000 transactions across 100 customers and 194 accounts between January 2020 and December 2025.

The headline finding isn't a single number, it's a gap: the bank moves $25M through a 95.3% reliable transaction pipeline, but two-thirds of its customer base is Suspended or Inactive, and nobody has connected those two facts yet. That's the problem this analysis exists to start closing.

KPI	Value
Total Transactions	**10,000**
Total Transaction Value	**$25,077,590.56**
Average Transaction Value	**$2,507.76**
Transaction Success Rate	**95.3%**
Transaction Failure Rate	**4.7%**
Total Customers	**100**
Active Customer %	**33%**
Suspended Customer %	**38%**
Inactive Customer %	**29%**
Average Transactions per Customer	**100**
Debit Share of Value	**60.3% ($15.12M)**
Credit Share of Value	**39.7% ($9.95M)**
Highest-Volume Channel	**Mobile (5,044 txns, $12.70M)**
Highest Failure-Rate Channel	**ATM (5.34%)**
Total Accounts	**194 (86 Open / 107 Closed)**

Objectives

The project was scoped to answer a defined set of business questions using SQL, validate patterns statistically, and surface the results in dashboards leadership can filter and drill into. Specifically:

- Quantify overall transaction volume, value, and average ticket size, and how they trend monthly and yearly.
- Break down performance by transaction type (Debit/Credit) and channel (Mobile/Web/ATM), including success and failure rates.
- Identify which customers, regions, account types, and products generate the most transaction value.
- Compare transaction activity across customer status (Active/Suspended/Inactive) and account status (Open/Closed).
- Test whether observed differences (Debit vs. Credit value, channel value, customer-status vs. channel usage) are statistically significant or within normal variation.
- Package the findings into dashboards and a written report that support channel investment, retention, and product decisions.

Dataset Description

The model is a standard star schema: a transaction fact table at the center, joined to customer (via account), product (via subcategory and category), and a date dimension. Full column-level detail is in the accompanying data_directory.md file; the summary below covers grain and scale.

Table	Grain	Rows	Key Columns
customer	1 row per customer	100	CustomerID (PK), Region, Status, JoinDate
account	1 row per account	194	AccountID (PK), CustomerID (FK), AccountType, Status, Balance
fact_transaction	1 row per transaction	10,000	TransactionID (PK), AccountID (FK), ProductID (FK), Amount, Type, Channel, Status
product	1 row per product	27	ProductID (PK), ProductSubcategoryID (FK)
subcategory	1 row per subcategory	9	ProductSubCategoryID (PK), ProductCategoryID (FK)
category	1 row per category	3	ProductCategoryID (PK)

Note on the schema diagram: the Data_Model.png file referenced alongside the other uploads wasn't actually present in the uploaded files (only an inline screenshot of it appeared in chat, not a saved file), so it isn't embedded here. Re-upload it if you want the visual ERD in the next version of this report — the table above carries the same relationships.

Data Quality

- Zero missing values in fact_transaction, product, subcategory, or category.
- ClosedDate in the account table is null only for the 86 Open accounts — consistent, no orphaned nulls.
- DimCustomerUSA.csv is an exact duplicate of DimCustomer.csv and should be dropped from the model to avoid double-counting if both are ever loaded.
- TransactionAmount is signed (positive/negative); every KPI in this report uses ABS(TransactionAmount) so inflows and outflows don't cancel each other out.

SQL Analysis

Eighteen business questions were answered directly in T-SQL against the Banking database (see Business_Solution.sql). The queries fall into five groups:

1. Overall Transaction Performance

- Total transactions, total value, and average value (Q1).
- Debit vs. Credit comparison (Q2), success/failure split (Q3).

2. Channel Performance

- Highest-volume and highest-value channel (Q4), highest-failure-rate channel (Q5), and the Debit/Credit mix by channel (Q6).

3. Time-Based Trends

- Month-by-month and year-by-year transaction volume and value (Q7, Q8), highest-value months (Q9), and month-over-month growth using a window function (Q10).

4. Customer, Account, and Region Performance

- Top customers by value and by count (Q11, Q14), activity by customer status (Q12), value by region (Q13), value by account type (Q15), and Open vs. Closed account activity (Q16).

5. Product Performance

- Value by product category via the category → subcategory → product join chain (Q17), and top individual products by value (Q18).

Results from these queries feed directly into the KPI table in the Executive Summary and the dashboard visuals later in this report. The full, runnable SQL is provided separately in Business_Solution.sql.

Python Exploratory Data Analysis

Before any statistical testing, the raw CSV exports were loaded and audited in Exploratory_Data_Analysis.ipynb. This stage is data-cleaning and validation, not insight generation — its job is to make sure the statistical analysis that follows is trustworthy.

What was checked

- Schema and dtypes for all five source tables, confirmed against expectations.
- Date fields parsed explicitly per table (customer DOB/JoinDate as dd/mm/yyyy, account dates and transaction dates as yyyy-mm-dd and mm/dd/yyyy respectively) — a reminder that the three date columns arrived in three different formats and needed separate parsing rules, not a single blanket conversion.
- Null counts per table — confirmed nulls in account.ClosedDate map exactly to Open accounts, and no other table carries unexpected nulls.
- Duplicate-row checks across all tables.
- Primary-key uniqueness validation for CustomerID, AccountID, TransactionID, and ProductID.
- Referential-integrity checks — every AccountID in the transaction table resolves to a valid account, and every CustomerID in the account table resolves to a valid customer.
- Descriptive summaries (describe()) across all tables to sanity-check ranges before modeling.

The cleaned tables were then loaded into SQL Server for the query-based analysis above. No records were dropped during this stage — the data passed its integrity checks.

Statistical Analysis

Statistical_Analysis.ipynb moves past description into testing — separating what looks like a pattern from what's actually significant at customer level (100 customers, aggregated from the 10,000 transactions).

Descriptive Analysis

Customer age averages 42.38 years (median 43). Customers make an average of 100 transactions each (median 97.5), generating an average total transaction value of $250,775.91 per customer (median $244,094.47), at an average per-transaction value of $2,503.48 (median $2,495.57). Mean and median are close across every metric, meaning customer activity is not heavily skewed at the aggregate level.

Distribution & Outlier Analysis

Using the IQR method at the customer level:

- Total transaction value per customer ranges ~$95,000–$470,000, with no IQR outliers — no single customer or handful of customers dominates total value.
- Transaction frequency ranges ~40–180 transactions per customer, also with no IQR outliers.
- Average transaction value is concentrated around $2,400–$2,650, with one customer falling below the lower IQR bound of $2,097.81 — a candidate for follow-up, not automatic removal, since an unusual customer pattern can be business-relevant rather than a data error.

Correlation Analysis

- Transaction frequency and total transaction value: r = 0.99 — an almost perfectly linear relationship. This is the single strongest signal in the dataset: customers who transact more often generate proportionally more total value.
- Customer age vs. transaction frequency: r = −0.06; age vs. total value: r = −0.06; age vs. average transaction value: r = 0.07. Age has essentially no relationship with financial activity.
- Transaction frequency vs. average transaction value: r = 0.06 — transacting more often doesn't mean transacting bigger.

Practical read: engagement frequency, not demographics, is what predicts customer value. Any retention or upsell targeting should be built around activity, not age.

Hypothesis Testing

Independent T-Test — Debit vs. Credit average value

H0: mean transaction value is equal for Debit and Credit. Debit averages $2,505.06, Credit averages $2,511.86 — a $6.80 gap. p = 0.8162, well above α = 0.05. Fail to reject H0: the gap is noise, not a real effect.

One-Way ANOVA — average value across Mobile / Web / ATM

H0: mean transaction value is equal across all three channels. Observed averages: Mobile $2,516.94, Web $2,474.49, ATM $2,534.81. F = 1.2554, p = 0.285. Fail to reject H0: channel doesn't meaningfully move average transaction size, even though ATM has the highest failure rate operationally.

Chi-Square Test — customer status vs. channel choice

H0: customer status and channel are independent. All three status groups (Active/Suspended/Inactive) favor Mobile most, but p = 0.2143 — fail to reject H0. Suspended and Inactive customers are not using different channels than Active ones; the retention problem isn't a channel-preference problem.

Confidence Interval

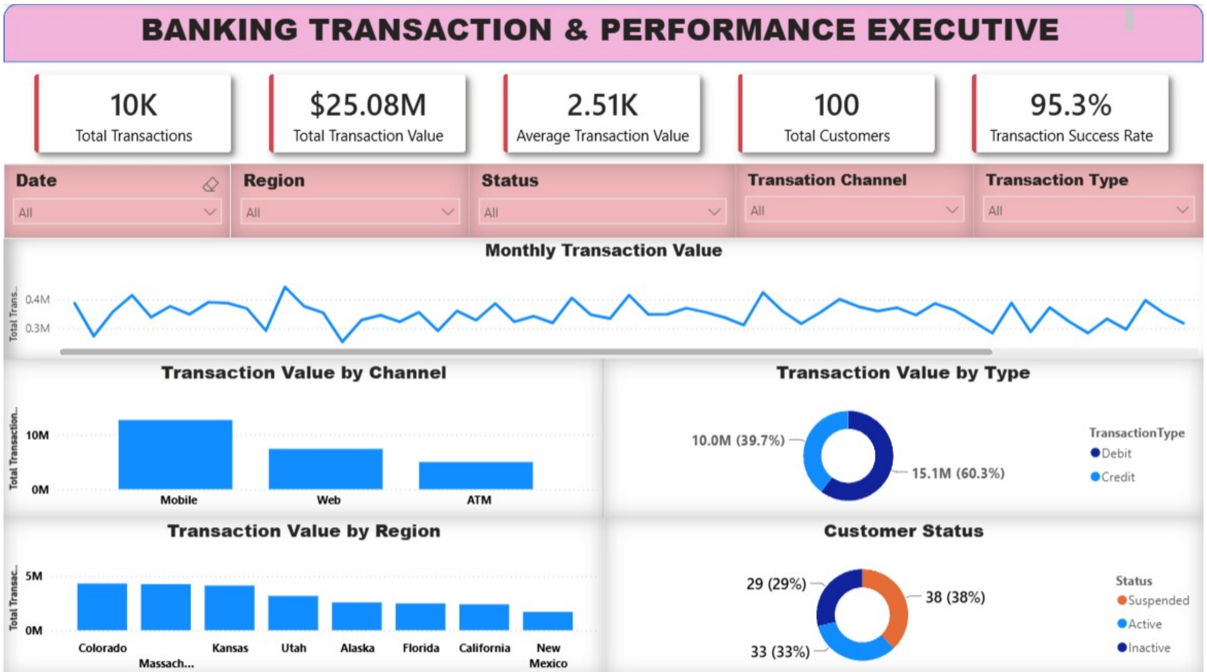
A 95% confidence interval was computed for the true mean transaction value using the t-distribution on all 10,000 transactions: $2,507.76 ± the interval, giving a range of $2,479.62 to $2,535.90. The channel-level averages calculated earlier (Mobile $2,516.94, Web $2,474.49, ATM $2,534.81) all fall inside or very close to this overall interval, which is consistent with — and reinforces — the ANOVA result that channel doesn't produce a statistically distinct average transaction value.

Key Statistical Findings

- Transaction frequency is the dominant driver of customer value (r = 0.99); age, transaction type, and channel are not.
- None of the three hypothesis tests found a statistically significant effect — Debit/Credit value, channel value, and customer-status/channel association are all consistent with random variation, not a real pattern.
- This means the 38% Suspended / 29% Inactive customer problem is not explained by channel behavior or transaction type — the cause sits elsewhere (service experience, product fit, or something not captured in this dataset) and needs separate investigation.

Dashboard Summaries

1. Banking Transaction & Performance Executive Overview

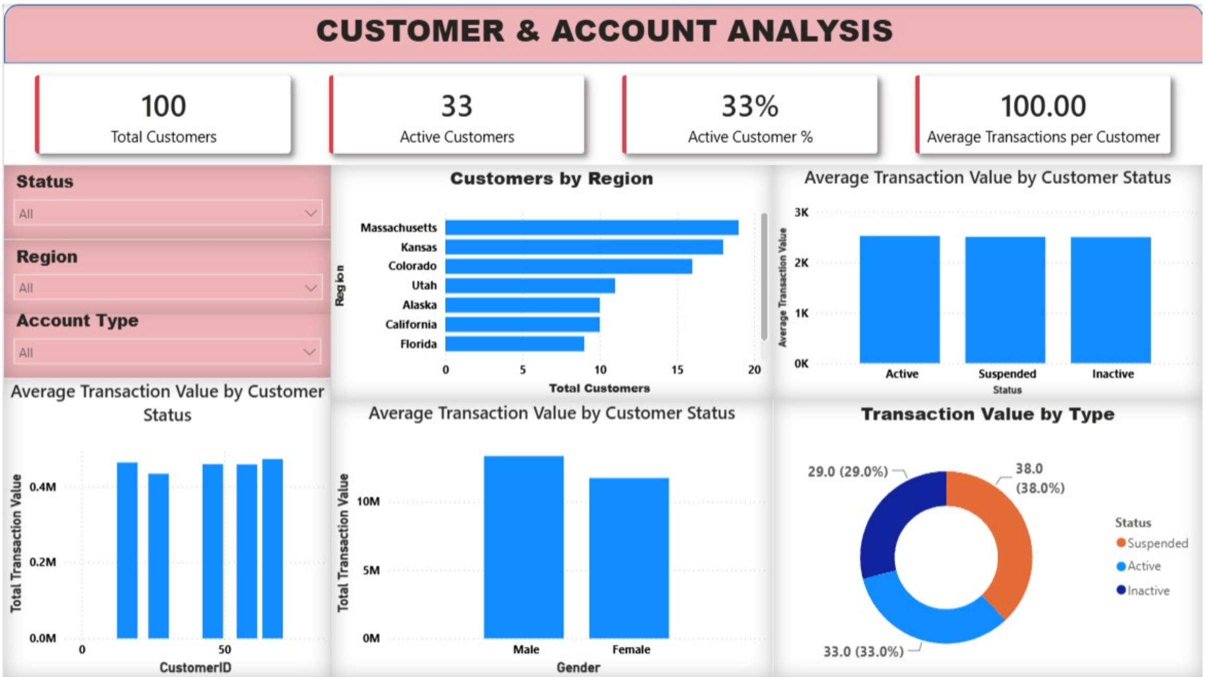


Executive Overview dashboard — filterable by Date, Region, Status, Channel, and Transaction Type.

Top-line KPIs: 10K transactions, $25.08M total value, $2.51K average value, 100 customers, 95.3% success rate. The monthly trend line shows activity oscillating between roughly $0.28M–$0.40M with no sustained directional trend across the period. Mobile leads channel value, and the customer-status donut confirms the 33/38/29 Active/Suspended/Inactive split at a glance.

One thing worth flagging directly: the Transaction Value by Type donut on this dashboard labels Debit at 39.7% ($10.0M) and Credit at 60.3% ($15.1M). Recalculating directly from the transaction-level data (with ABS applied) gives the opposite assignment — Credit is 39.7% ($9.95M) and Debit is 60.3% ($15.12M). The percentages are correct; the Debit/Credit legend labels appear to be swapped in this visual. Worth a quick fix in the .pbix before this goes to stakeholders.

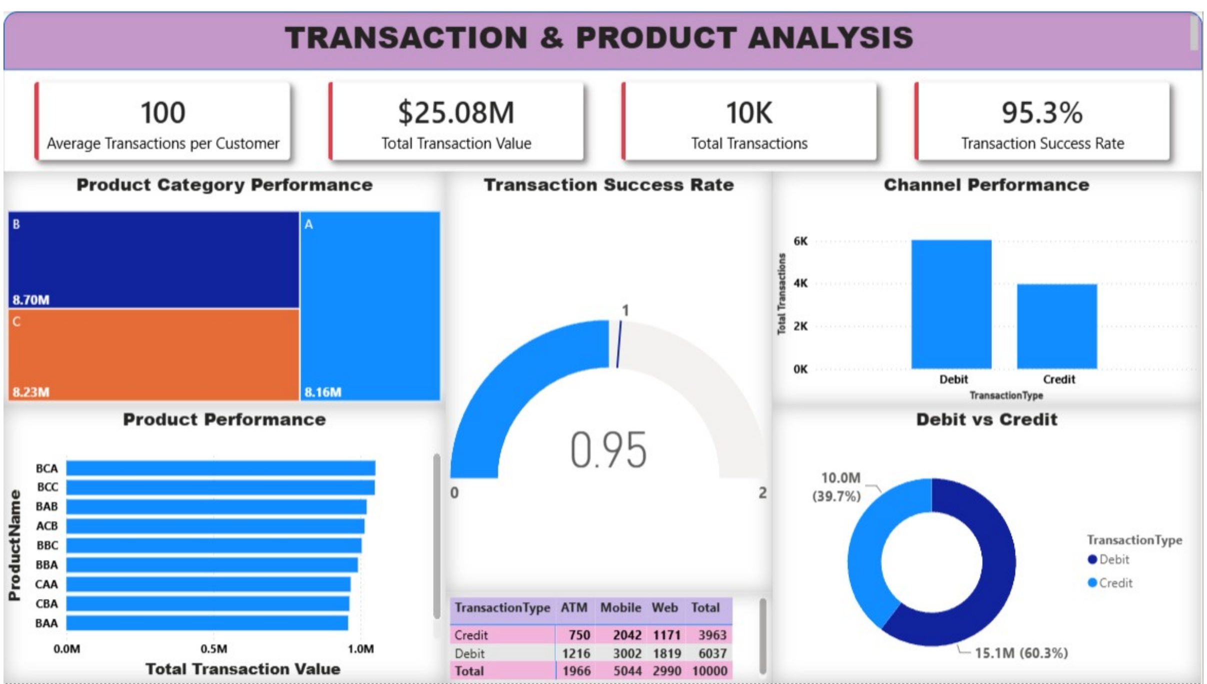
2. Customer & Account Analysis



Customer & Account Analysis dashboard — filterable by Status, Region, and Account Type.

100 total customers, 33 Active (33%), 100 average transactions per customer. Massachusetts, Kansas, and Colorado lead by customer count. Average transaction value is fairly flat across Active, Suspended, and Inactive customers — reinforcing the statistical finding that status doesn't correlate with transaction behavior. The Male/Female split in total transaction value skews toward Male customers, which is worth a follow-up cut once more demographic detail is available.

3. Transaction & Product Analysis



Transaction & Product Analysis dashboard — filterable and drillable by product and transaction type.

Product category performance is close to even across A ($8.16M), B ($8.70M), C ($8.23M) — no single category dominates. The transaction success gauge confirms 0.95 (95%) against a 0–2 scale visual. Channel performance by transaction type shows Debit consistently outweighing Credit in count across ATM, Mobile, and Web, and the ATM/Mobile/Web breakdown table (1,966 / 5,044 / 2,990, totaling 10,000) ties out exactly to the fact table.

Key Business Insights

- The bank processes $25.08M across 10,000 transactions at a 95.3% success rate — the core transaction pipeline is healthy on its own terms.
- Two-thirds of the customer base (67%) is Suspended or Inactive, and this is the single biggest business risk in the dataset. It is not explained by channel choice, transaction type, or customer age — those were all statistically ruled out.
- Transaction frequency (not age, not channel, not type) is what actually predicts customer value (r = 0.99). Any model built to predict or grow customer value should be built on engagement/frequency signals.
- ATM has the highest failure rate (5.34%) of the three channels, more than a full point above Web (3.95%). It's a smaller channel by volume (1,966 of 10,000 transactions) but a disproportionate share of the failure count.
- Debit transactions outnumber Credit by count (6,037 vs. 3,963) and by value (60.3% vs. 39.7%) — the bank is net-debit-heavy, which matters for liquidity and cash-flow planning.
- No product category or individual customer dominates the portfolio — value is broadly distributed, which is a resilience positive (no single point of failure) but also means there's no obvious standout product to double down on without further segmentation.

Business Recommendations

- Launch a dedicated investigation into the 67% Suspended/Inactive customer base. Since channel, transaction type, and age are all statistically ruled out as causes, the next step is looking at service history, product fit, or account-level friction (e.g., the 107 Closed accounts) — data this analysis didn't have access to.
- Fix the ATM channel's failure rate before investing further in ATM infrastructure — at 5.34% it's the weakest link operationally, even though it doesn't affect average transaction value.
- Build customer engagement/frequency into any retention or growth model, given its near-perfect correlation with total transaction value; deprioritize age-based segmentation, which showed no relationship.
- Correct the Debit/Credit legend swap on the Executive Overview dashboard before it's shared further — the underlying numbers are right, the labels aren't.
- Consolidate DimCustomer.csv and DimCustomerUSA.csv in the source pipeline to prevent an accidental double-count if both ever get loaded into the same model.

Future Improvements

- Bring in the fields this dataset doesn't have — service/complaint history, product usage detail, fee data — to actually explain the Suspended/Inactive gap rather than just ruling out what it isn't.
- Document RegistrationID (values 1–3) in the account table; it currently has no lookup table or definition anywhere in the source files.
- Extend the statistical testing to account type and account status (Open/Closed) against transaction value, which wasn't in the original hypothesis set.
- Automate the SQL-to-dashboard refresh so the KPI table in this report and the .pbix visuals never drift out of sync.
- Re-embed the schema/ERD image once Data_Model.png is actually uploaded, replacing the text-table substitute used in this version.

Conclusion

The transaction side of this business is in good shape: high volume, high reliability, no single point of concentration risk. The customer side is not — a 33% Active rate is the number that should be driving the next quarter's priorities, and this analysis has already done the useful work of ruling out the easy explanations (channel, transaction type, age). What's left is the harder, more specific investigation into why customers are leaving or getting suspended, and that requires data this report didn't have. Until that's answered, treat every other recommendation here as secondary to fixing that number.